

Experiment-3A

Aim: WORKING WITH TEMPLATE ENGINE.

Description:

A template engine is a tool that allows you to generate HTML dynamically on the server by combining:

HTML (structure)

Data (from server / database)

Logic (conditions & loops)

Instead of sending plain HTML files, the server sends processed HTML.

Program:

```
const ejs=require('ejs');
const express=require('express');
const app=express();
const port=5000;
app.set('view engine','ejs');
app.get('/time',(req,res)=>{
  res.render('time');
})
app.get('/login',(req,res)=>{
  res.render('login');
})
app.get('/register',(req,res)=>{
  res.render('register');
})
app.listen(port,()=>{
  console.log('Server Started 5000 port');
})
```

Login.ejs

```
<html>
<head>
<title>Login</title>
<style>
body{
```

```

margin:0;height:100vh;display:flex;align-items:center;justify-content:center;
background:linear-gradient(135deg,#020617,#1e3a8a);font-family:Arial;
}
.card{
background:#fff;padding:30px;width:320px;border-radius:12px;
box-shadow:0 20px 40px rgba(0,0,0,.3);
}
h2{text-align:center;margin-bottom:20px}
input{width:100%;padding:10px;margin-bottom:12px;border:1px solid #ccc}
button{width:100%;padding:10px;background:#1e3a8a;color:#fff;border:none}
a{display:block;text-align:center;margin-top:12px;color:#1e3a8a;text-decoration:none}
</style>
</head>
<body>
<div class="card">
  <h2>Login</h2>
  <input type="email" placeholder="Email">
  <input type="password" placeholder="Password">
  <button>Login</button>
  <a href="register.ejs">New user? Register here</a>
  <a href="time.ejs">Go to time table page</a>
</div>
</body>
</html>

```

Register.ejs

```

<html>
<head>
<title>Register</title>
<style>
a{
display:block;text-align:center;margin-top:10px;text-decoration:none;
}
</style>
</head>
<body>
<form>
<h2 align="center">Register</h2>
<label>Name</label>

```

```

<input type="text">
<label>Email</label>
<input type="email">
<label>Phone</label>
<input type="tel">
<label>Address</label>
<textarea></textarea>
<label>Password</label>
<input type="password">
<button>Register</button>
<a href="login.ejs">Login</a>
</form>
</body>
</html>

```

Time.ejs

```

<!DOCTYPE html>
<html>
<head>
<title>Class Timetable</title>
<style>
body{
  font-family:Arial; background:#f1f5f9; padding:20px;
}
h2{
  text-align:center;
}
table{
  width:100%; border-collapse:collapse; background:#fff;
}
</style>
</head>
<body>
<h2>Section A – Class Timetable</h2>
<p><strong>Class Teacher:</strong> Mrs. PPS Katyayini
&nbsp;&nbsp;&nbsp;<strong>Academic Coordinator:</strong> Dr. A. Aparna</p>
<table>
<tr>
<th>Day</th>

```

```

<th>9:00–9:50</th>
<th>9:50–10:40</th>
<th>10:40–11:30</th>
<th>11:30–12:20</th>
<th>12:20–1:20</th>
<th>1:20–2:10</th>
<th>2:10–3:00</th>
<th>3:00–3:50</th>
<th>3:50–4:50</th>
</tr>
<tr>
<td>Mon</td>
<td>CNS</td>
<td>IPR</td>
<td>CC</td>
<td>ML</td>
<td rowspan="6">LUNCH</td>
<td>FSD-II</td>
<td colspan="3">CC LAB</td>
</tr>
<tr>
<td>Tue</td>
<td>PE-III</td>
<td>CNS</td>
<td>FSD-II</td>
<td>LIB</td>
<td>CC</td>
<td>CNS</td>
<td>CD</td>
<td>ML</td>
</tr>
<tr>
<td>Wed</td>
<td>CC</td>
<td>ML</td>
<td>CD</td>
<td>PE-III</td>
<td colspan="3">FSD-II LAB</td>

```

```

    <td>COUN / ASSOC</td>
</tr>
<tr>
    <td>Thu</td>
    <td colspan="2">OE-II</td>
    <td>PE-III</td>
    <td>H/M ML</td>
    <td>IPR</td>
    <td>CC</td>
    <td>CD</td>
    <td>CNS</td>
</tr>
<tr>
    <td>Fri</td>
    <td>CD</td>
    <td>CNS</td>
    <td>ML</td>
    <td>H/M CC</td>
    <td colspan="2">OE-II</td>
    <td>PE-III</td>
    <td>H/M CD</td>
</tr>
<tr>
    <td>Sat</td>
    <td colspan="3">CNS LAB</td>
    <td>H/M CNS</td>
    <td>CD</td>
    <td>ML</td>
    <td>CC</td>
    <td></td>
</tr>
</table>
</body>
</html>

```

Output:

```

PROBLEMS  OUTPUT  TERMINAL  PORTS  POSTMAN CONSOLE  DEBUG CONSOLE

PS D:\24485A0503\Experiment-1> node app
Server running on http://localhost:3000

```

Browser window: Login

URL: localhost:5000/login

Login

Login

New user? Register here

Go to time table page

Browser window: Register

URL: localhost:5000/register

Register

Register

Login

Browser window: Class Timetable

URL: localhost:5000/time

Section A – Class Timetable

Class Teacher: Mrs. PPS Katayayini Academic Coordinator: Dr. A. Aparna

Day	9:00-9:50	9:50-10:40	10:40-11:30	11:30-12:20	12:20-1:20	1:20-2:10	2:10-3:00	3:00-3:50	3:50-4:50
Mon	CNS	IPR	CC	ML	LUNCH	FSD-II	CC LAB		
Tue	PE-III	CNS	FSD-II	LIB		CC	CNS	CD	ML
Wed	CC	ML	CD	PE-III		FSD-II LAB			COUN / ASSOC
Thu	OE-II		PE-III	H/M ML		IPR	CC	CD	CNS
Fri	CD	CNS	ML	H/M CC		OE-II		PE-III	H/M CD
Sat	CNS LAB			H/M CNS		CD	ML	CC	

Experiment-3B

Aim: WORKING WITH FORM DATA

Description:

Working with form data is the process of capturing user input through HTML forms and handling it on the server side to perform required operations such as validation, processing, storage, or response generation. Forms act as a bridge between the user interface and the server logic in a web application.

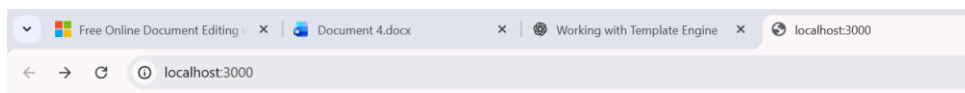
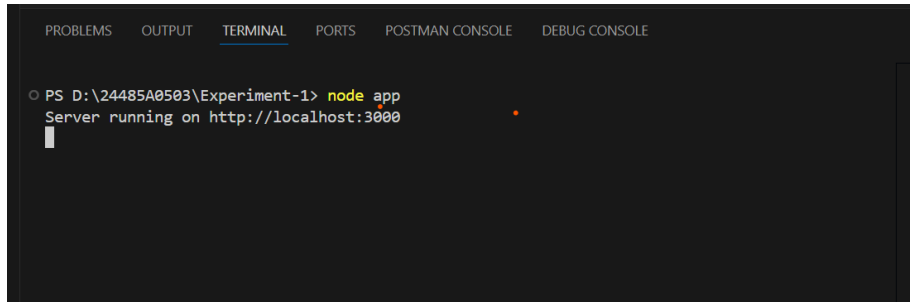
When a user submits a form, the browser sends the entered data to the server using an HTTP request. The server receives this data, extracts the values, verifies them, and performs appropriate actions like saving to a database or displaying results dynamically.

Program:

```
const express = require('express');
const app = express();
// Middleware to read form data
app.use(express.urlencoded({ extended: true }));
// Home page with form
app.get('/', (req, res) => {
  res.send(`
    <h2>Student Registration Form</h2>
    <form method="POST" action="/submit">
      Name: <input type="text" name="name" /><br><br>
      Email: <input type="email" name="email" /><br><br>
      Age: <input type="number" name="age" /><br><br>
      <input type="submit" value="Submit" />
    </form>
  `);
});
// Handle form submission
app.post('/submit', (req, res) => {
  const { name, email, age } = req.body;
  res.send(`
    <h2>Form Data Received</h2>
    Name: ${name}<br>
    Email: ${email}<br>
    Age: ${age}
  `);
});
```

```
`);  
});  
// Server  
app.listen(3000, () => {  
  console.log('Server running on http://localhost:3000');  
});
```

Output

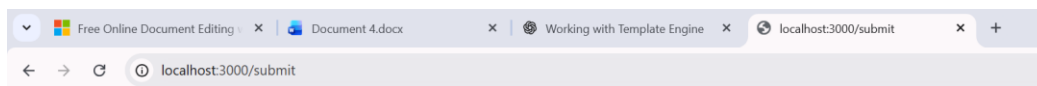


Student Registration Form

Name:

Email:

Age:



Form Data Received

Name: Sai
Email: sai@gmail.com
Age: 20